

# BISWARANJAN SAHOO



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## PROFESSIONAL SUMMARY

Innovative and results driven Engineering Student with Technical Mindset with a proven track record of bridging hardware architecture and software development. Experienced in leading end-to-end technical deployments from edge AI and autonomous drone systems to cloud-integrated solar energy optimization. Good foundation in machine learning operations, cloud infrastructure, and regulatory compliance, seeking to leverage a diverse technical skill set in a dynamic engineering internship.

## EDUCATION

**Biju Patnaik University of Technology (BPUT)** | Rourkela, Odisha, India

Bachelor of Technology in Computer Science & Engineering

Expected Graduation: Jul, 2027

## PROFESSIONAL EXPERIENCE

### Chief Technical & Compliance Lead

Mar, 2025 – Present

AkashVahini Private Limited | Incubated at FTBI, NIT Rourkela | Rourkela, Odisha (ROC Bhubaneswar)

- Directed the technical architecture and prototype development of advanced aerial surveillance and rescue drone systems, integrating carbon fiber frames, high-efficiency motors, and custom flight controllers.
- Successfully navigated the startup through competitive selection for the prestigious MeitY GENESIS – Entrepreneur-in-Residence (EIR) 2.0 program.
- Oversaw comprehensive company compliance and regulatory documentation while aligning technical deliverables with overarching business goals.

## TECHNICAL PROJECTS

### SouryaNova | Cloud-Integrated Solar Optimization System

- Designed a dual-optimization solar tracking platform utilizing hardware sensors (LDRs, MPPT algorithms) for physical panel alignment.
- Developed "SouryaNova Brain AI" leveraging AWS Bedrock, AWS Lambda, and S3 databases for advanced software monitoring and efficiency analysis.

## Autonomous Drone & Edge AI Infrastructure

- ❑ Configured and deployed a cutting-edge 16 GB Raspberry Pi 5 paired with a Hailo AI accelerator for on-device edge computing.
- ❑ Implemented sophisticated computer vision models, depth mapping for enhanced autonomous navigation for robotic and drone platforms.

## AI-Driven Online Exam Management System

- ❑ Built a comprehensive automated proctoring solution integrating real-time computer vision to detect behavioral anomalies and maintain examination integrity.

## Data Analytics and ML Training | Central Tool Room & Training Centre (MSME), Government of India | Bhubaneswar, Odisha

- ❑ **Completed Govt. of India (MSME) AI/ML internship**, deploying supervised and unsupervised models for Iris Flower classification, attendance system and predictive machine breakdown.
- ❑ **Implemented Natural Language Processing (NLP)** pipelines utilizing TF-IDF and engineered deep learning architectures via Keras.
- ❑ **Developed a capstone computer vision model** to accurately detect and classify German traffic signboards.

## TECHNICAL SKILLS

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|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Coding Languages:</b>        | Python, C, Java                                                                                                                                                                                                          |
| <b>Scripting Languages:</b>     | JavaScript (React, ShadCN), TypeScript                                                                                                                                                                                   |
| <b>Environments:</b>            | Node.js                                                                                                                                                                                                                  |
| <b>Cloud &amp; Databases:</b>   | AWS (Bedrock, Lambda, S3), Drizzle ORM                                                                                                                                                                                   |
| <b>Hardware &amp; Embedded:</b> | Raspberry Pi OS, Drone Flight Controllers, Solar Tracking Hardware (LDR, MPPT)                                                                                                                                           |
| <b>Systems &amp; Dev Tools:</b> | Linux (Ubuntu, Debian), Command-Line Administration                                                                                                                                                                      |
| <b>Artificial Intelligence:</b> | Edge AI computing, Computer Vision, MLOps, Hailo AI integration, Natural Language Processing (NLP), Supervised & Unsupervised Learning, Data Analytics (Pandas, NumPy, Keras, TensorFlow, Scikit Learn, Matplotlib etc.) |

## ACHIEVEMENTS & LEADERSHIP

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**2<sup>nd</sup>, VSSUT Hack for Tomorrow 2026** — 3 days long hackathon related to hardware and software integration leveraging the use of IoT and Explainable AI.

**Finalist, Trithon 2026 Hackathon** — Recognized for technical excellence and innovation in developing the SuryaNova solar energy project.

**Event Organizer** — Acted as a liaison for BPUT students; successfully coordinated and drove participation for the BPUT x Google Developers Group Ranchi "DevSpark" technology event.